

08.1

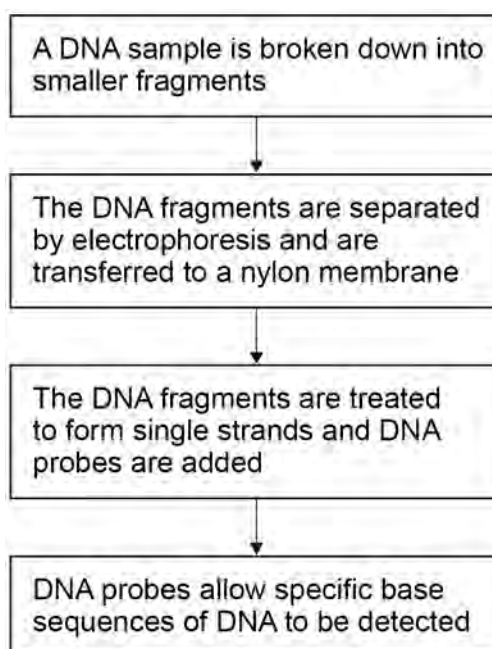
What is a DNA probe?

[2 marks]

DNA probes are used to detect specific base sequences of DNA.

The process is shown in **Figure 6**.

Figure 6



08.2

Describe how the DNA is broken down into smaller fragments.

[2 marks]

0 8 . 3

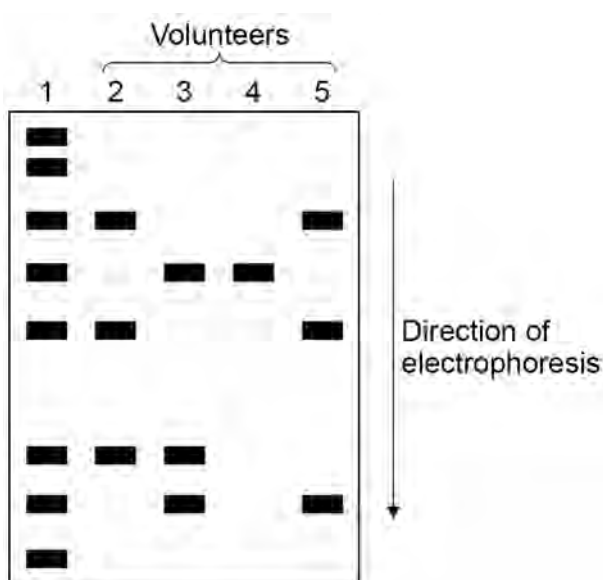
The DNA on the nylon membrane is treated to form single strands. Explain why.

[1 mark]

A scientist used DNA probes and electrophoresis to screen four volunteers for five different viral DNA fragments.

Figure 7 shows the results the scientist obtained. The lanes numbered 2 to 5 represent the four volunteers.

Figure 7



0 8 . 4

Lane 1 of **Figure 7** enabled the size of the different viral fragments to be determined.

Suggest and explain how.

[2 marks]

Turn over ►

The lengths of the viral DNA fragments were:

- 600 base pairs
- 250 base pairs
- 535 base pairs
- 300 base pairs
- 500 base pairs.

0 8 . 5

Which volunteers had at least one of the viral DNA fragments with 250 base pairs or 535 base pairs?

[1 mark]

8

